



Session 2: Bayesian Survival Analysis

Proportional Hazards Models and Competing Risks

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Session Overview

Survival Analysis Recap

Model 1: Exponential

Model 2: Weibull

Model 3: Gompertz

Model 4: Competing Risks

Model Comparison

Summary

References

Survival Analysis Recap

Key Quantities in Survival Analysis

Survival function:

$$S(t) = P(T > t) = 1 - F(t)$$

Probability of surviving past time t .

Hazard function:

$$h(t) = \lim_{\delta \rightarrow 0} \frac{P(t \leq T < t + \delta \mid T \geq t)}{\delta}$$

Instantaneous risk of event at time t .

Cumulative hazard:

$$H(t) = \int_0^t h(s) ds$$

Key relationships:

$$S(t) = \exp(-H(t))$$

$$f(t) = h(t) S(t)$$

$$H(t) = -\log S(t)$$

Right censoring

Subject i contributes:

- **Event** ($\delta_i = 1$): $f(t_i)$
- **Censored** ($\delta_i = 0$): $S(t_i)$

$$\log L = \sum_i \delta_i \log h(t_i) - H(t_i)$$

The PH assumption

$$h(t \mid \mathbf{x}_i) = h_0(t) \cdot \exp(\mathbf{x}_i' \boldsymbol{\gamma})$$

Covariates **multiply** the baseline hazard $h_0(t)$ by a constant factor that does not depend on time.

$\exp(\gamma_k) = \textbf{hazard ratio (HR)}$ for a 1-unit increase in x_k .

Why PH models?

- HRs are directly interpretable as relative risks
- Natural extension to joint models (Session 3)
- Bayesian PH: closed-form $H(t)$ for Exponential, Weibull, Gompertz
- Posterior of $\exp(\boldsymbol{\gamma})$ gives full uncertainty

Three PH families (this session)

1. **Exponential:** $h_0(t) = \lambda$; constant
2. **Weibull:** $h_0(t) \propto t^{\alpha-1}$; monotone
3. **Gompertz:** $h_0(t) = \lambda e^{\rho t}$; exponential growth/decay

Hazard Shape Taxonomy: PH Models

Model	$h_0(t)$	Hazard shape	$H_0(t)$ tractable?	Key param.
Exponential	λ	Constant flat	Yes: λt	$\lambda > 0$
Weibull	$\alpha \lambda t^{\alpha-1}$	Monotone incr/decr	Yes: $(\lambda t)^\alpha$	α : shape
Gompertz	$\lambda e^{\rho t}$	Exponential incr/decr	Yes: $\frac{\lambda}{\rho}(e^{\rho t} - 1)$	ρ : rate
Competing risks	Per-cause Gompertz	Cause-specific	Yes (per cause)	λ_k, ρ_k

All three models share:

- PH structure: $h(t|\mathbf{x}) = h_0(t) \exp(\mathbf{x}'\boldsymbol{\gamma})$
- Analytically tractable $H_0(t)$; no quadrature needed in Stan
- Hazard ratios $\exp(\gamma_k)$ as primary summary

PBC context

Disease progression worsens over time: expect **increasing hazard**.

Weibull ($\hat{\alpha} \approx 1.45$) and Gompertz ($\hat{\rho} = 0.086$, $P(\rho > 0) = 0.998$) both confirm this.

Model 1: Exponential (Constant Hazard)

Model specification

$$h(t \mid \mathbf{x}_i) = \lambda \exp(\mathbf{x}_i' \boldsymbol{\gamma}), \quad H(t \mid \mathbf{x}_i) = \lambda t \exp(\mathbf{x}_i' \boldsymbol{\gamma})$$

Simplest PH model: baseline hazard is **constant** at λ ; the event rate does not change with time.

Stan log-likelihood contribution:

$$\delta_i = 1 \text{ (event)} : \quad \log \lambda + \mathbf{x}_i' \boldsymbol{\gamma} - H_i$$

$$\delta_i = 0 \text{ (censored)} : \quad -H_i$$

Priors (weakly informative):

$$\lambda \sim \text{Gamma}(1, 1) \quad [\text{mean} = 1]$$

$$\boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Special properties

- **Memoryless:**
 $P(T > s + t \mid T > s) = P(T > t)$
- **Median survival:** $t_{1/2} = \log 2 / \lambda$
- Special case of Weibull ($\alpha = 1$) and Gompertz ($\rho = 0$)

Exponential PH Model — Structure

```
1 // surv_exponential.stan --- Constant hazard PH model
2
3 data {
4   int<lower=1> N; // number of subjects
5   vector<lower=0>[N] t_event; // observed times
6   array[N] int<lower=0,1> event; // 1=event, 0=censored
7   int<lower=1> P; // number of covariates
8   matrix[N, P] X; // design matrix
9 }
10
11 parameters {
12   real<lower=0> lambda; // baseline hazard
13   vector[P] gamma; // log hazard ratios
14 }
15
16 transformed parameters {
17   vector[N] log_lambda_x;
18   log_lambda_x = log(lambda) + X * gamma;
19 }
```

Key idea:

$$h_i(t) = \lambda \exp(X_i \gamma), \quad H_i(t) = t \lambda \exp(X_i \gamma)$$

Exponential PH Model: Likelihood

```
1  model {  
2    // Priors  
3    lambda ~ gamma(1, 1);  
4    gamma ~ normal(0, 1);  
5  
6    // Likelihood  
7    for (i in 1:N) {  
8      real H_i = t_event[i] * exp(log_lambda_x[i]);  
9  
10     if (event[i] == 1)  
11       target += log_lambda_x[i] - H_i;    // event  
12     else  
13       target += -H_i;                      // censored  
14   }  
15 }
```

Mathematical form:

$$\ell_i = \begin{cases} \log(\lambda \exp(X_i \gamma)) - H_i(t) & \text{if event} \\ -H_i(t) & \text{if censored} \end{cases}$$

Exponential PH Model — Outputs and Diagnostics

```
1 generated quantities {
2   vector[P] HR = exp(gamma);           // hazard ratios
3   real t_median_ref = log(2.0) / lambda; // baseline median
4
5   vector[N] log_lik;
6   vector[N] cox_snell;
7
8   for (i in 1:N) {
9     real H_i = t_event[i] * exp(log_lambda_x[i]);
10
11     log_lik[i] = (event[i] == 1)
12       ? log_lambda_x[i] - H_i
13       : -H_i;
14
15     cox_snell[i] = H_i;
16   }
17 }
```

Exponential: Posterior Summary

```
1 fit_exp$summary(c("lambda", "gamma", "HR", "t_median_ref")) |>
2   select(variable, mean, sd, q5, q95, rhat, ess_bulk)
3
4 # variable      mean      sd      q5      q95    rhat  ess_bulk
5 # lambda        0.069  0.006   0.057   0.082    1.00   3942
6 # gamma[1]     -0.116  0.163  -0.390   0.151    1.00   3817 (drug)
7 # gamma[2]      0.190  0.052   0.103   0.277    1.00   4103 (age/10yr)
8 # HR[1]         0.890  0.146   0.677   1.163    1.00   3817 (drug)
9 # HR[2]         1.209  0.063   1.108   1.319    1.00   4103 (age/10yr)
10 # t_median_ref 10.04   0.892   8.45  12.10    1.00   3942
```

- $\hat{\lambda} = 0.069$: approx. 6.9 events per 100 person-years
- Median survival ≈ 10.0 years for reference patient (placebo, mean age)
- Drug HR = 0.89 [0.68, 1.16]: modest protective effect, not significant
- Age HR = 1.21 per 10 yr [1.11, 1.32]: older patients clearly at higher risk
- $\hat{R} < 1.01$, ESS > 3800, zero divergences

Exponential: Comparison with Rizopoulos / coxph()

Rizopoulos (2012, Ch. 3) uses a Cox PH model on the same PBC data:

```
1 library(survival)
2 cox_fit <- coxph(Surv(t_event, event) ~ trt + age_per10yr,
3                 data = pbc_surv_df)
4 # coef      exp(coef) se(coef)      z      p
5 # trt          -0.135    0.874    0.180  -0.75  0.45
6 # age_per10yr  0.183    1.201    0.049   3.73  0.00
```

Covariate	Rizopoulos Cox HR	Stan Exp HR	Stan 90% CI
Drug (D-penicillamine)	0.87	0.89	[0.68, 1.16]
Age (per 10 yr)	1.20	1.21	[1.11, 1.32]

Agreement validates Stan; Exponential adds interpretability

HRs agree. Bayesian Exponential adds: $P(\lambda \mid \text{data})$, median survival CI, LOO diagnostic — and extends to the joint model (Session 3).

Model 2: Weibull

Model specification

$$h_0(t) = \alpha e^{-\alpha\mu} t^{\alpha-1}, \quad H_0(t) = \left(\frac{t}{e^\mu} \right)^\alpha$$

$$h(t \mid \mathbf{x}_i) = h_0(t) \exp(\mathbf{x}_i' \boldsymbol{\gamma}), \quad H(t \mid \mathbf{x}_i) = \left(\frac{t}{e^{\mu + \mathbf{x}_i' \boldsymbol{\gamma}/\alpha}} \right)^\alpha$$

Shape α controls monotone hazard direction; scale e^μ sets the time horizon.

Shape interpretation:

- $\alpha = 1$: reduces to Exponential (constant hazard)
- $\alpha > 1$: hazard **increases** over time (PBC, cancer, ageing)
- $\alpha < 1$: hazard decreases (bathtub hazard)

Priors:

$$\alpha \sim \text{Gamma}(2, 1) \quad [\mathbb{E}[\alpha] = 2]$$

$$\mu \sim \mathcal{N}(2, 4) \quad [\text{log-scale}]$$

$$\boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\text{Median survival: } t_{1/2} = e^\mu \cdot (\log 2)^{1/\alpha}$$

Weibull PH Model — Structure

```
1  data {  
2    int<lower=1> N;  
3    vector<lower=0>[N] t_event;  
4    array[N] int<lower=0,1> event;  
5    int<lower=1> P;  
6    matrix[N, P] X;  
7  }  
8  parameters {  
9    real<lower=0> alpha;    // shape  
10   real mu;                // log-scale  
11   vector[P] gamma;        // log hazard ratios  
12 }  
13 transformed parameters {  
14   vector[N] xgamma = X * gamma;  
15 }
```

$$h_i(t) = \alpha t^{\alpha-1} \exp(-\alpha\mu + X_i\gamma)$$

$$H_i(t) = \left(\frac{t}{\exp(\mu + X_i\gamma/\alpha)} \right)^\alpha$$

Weibull PH Model: Likelihood

```
1   transformed parameters {
2     vector[N] H;
3     for (i in 1:N)
4       H[i] = pow(t_event[i] / exp(mu + xgamma[i]/alpha), alpha);
5   }
6   model {
7     alpha ~ gamma(2, 1);
8     mu     ~ normal(2, 2);
9     gamma ~ normal(0, 1);
10    for (i in 1:N) {
11      if (event[i] == 1)
12        target += log(alpha) - alpha*mu
13        + (alpha-1)*log(t_event[i])
14        + xgamma[i] - H[i];
15      else
16        target += -H[i];}
17  }
```

$$\ell_i = \begin{cases} \log h_i(t_i) - H_i(t_i), & \text{event} \\ -H_i(t_i), & \text{censored} \end{cases}$$

Weibull PH Model: Outputs and Diagnostics

```
1 generated quantities {
2   vector[P] HR = exp(gamma);
3
4   vector[N] log_lik;
5   vector[N] cox_snell;
6
7   for (i in 1:N) {
8     log_lik[i] = (event[i] == 1) ?
9     log(alpha) - alpha*mu + (alpha-1)*log(t_event[i])
10    + xgamma[i] - H[i] : -H[i];
11
12    cox_snell[i] = H[i];
13  }
14 }
```

Weibull; Posterior Summary

```
1 fit_wb$summary(c("alpha", "mu", "HR")) |>
2   select(variable, mean, sd, q5, q95, rhat, ess_bulk)
3
4 # variable      mean      sd      q5      q95    rhat  ess_bulk
5 # alpha        1.448    0.113    1.230    1.672    1.00    3612
6 # mu           2.514    0.118    2.322    2.700    1.00    3444
7 # HR [1]        0.872    0.139    0.636    1.186    1.00    3590  (drug)
8 # HR [2]        1.228    0.085    1.069    1.409    1.00    3751  (age/10yr)
```

Key findings

- $\hat{\alpha} = 1.45$ [1.23, 1.67]: $P(\alpha > 1) > 0.999$; increasing hazard confirmed
- $e^{\hat{\mu}} = e^{2.51} = 12.3$ yr: scale of PBC progression
- Median survival ≈ 10.2 yr [8.7, 12.0] for reference group
- $\hat{R} < 1.003$, 0 divergences

$\alpha = 1$ Bayesian test

The posterior of α lies entirely above 1.0:

$$P(\alpha > 1 \mid \mathbf{y}) > 0.999$$

Strong evidence that PBC hazard increases monotonically with time.

Weibull: Comparison with Rizopoulos / coxph()

```
1 # Frequentist Weibull via flexsurv
2 library(flexsurv)
3 fsurv_wb <- flexsurvreg(Surv(t_event, event) ~ trt + age_per10yr,
4                         dist = "weibull", data = pbc_surv_df)
5 exp(coef(fsurv_wb))
6 # shape = 1.449    scale = 12.28    trt = 0.873    age_per10yr = 1.231
```

Parameter	flexsurv MLE	Stan Weibull (median)	Stan 90% CI
α (shape)	1.449	1.448	[1.23, 1.67]
e^μ (scale, yr)	12.28	12.3	[10.2, 14.8]
Drug HR	0.873	0.872	[0.64, 1.19]
Age HR (per 10 yr)	1.231	1.228	[1.07, 1.41]

Near-perfect agreement

Stan posterior medians match flexsurv MLE. Bayesian adds posterior over α enabling direct $P(\alpha > 1)$ test without asymptotic assumptions.

Model 3: Gompertz

Model

$$h(t \mid \mathbf{x}_i) = \lambda e^{\rho t} \exp(\mathbf{x}_i' \boldsymbol{\gamma})$$

$$H(t \mid \mathbf{x}_i) = \frac{\lambda}{\rho} (e^{\rho t} - 1) \exp(\mathbf{x}_i' \boldsymbol{\gamma}) \quad [\rho \neq 0]$$

Priors:

- $\rho > 0$: hazard **increases exponentially** with time
- $\rho < 0$: hazard decreases exponentially with time
- $\rho = 0$: reduces to exponential (guard in Stan: `fabs(rho) > 1e-6`)
- Originally proposed to model human mortality

$$\lambda \sim \text{Gamma}(1, 1)$$

$$\rho \sim \mathcal{N}(0, 1) \quad (\text{allows } \pm)$$

$$\boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Key generated quantity:

$P(\rho > 0 \mid \mathbf{y})$ = posterior probability that hazard increases.

Gompertz PH Model: Structure

```
1  data {  
2    int<lower=1> N;  
3    vector<lower=0>[N] t_event;  
4    array[N] int<lower=0,1> event;  
5    int<lower=1> P;  
6    matrix[N, P] X;  
7  }  
8  
9  parameters {  
10    real<lower=0> lambda;    // initial baseline hazard  
11    real rho;                // hazard growth rate  
12    vector[P] gamma;        // log hazard ratios  
13  }  
14  transformed parameters {  
15    vector[N] xgamma = X * gamma;  
16  }
```

$$h_i(t) = \lambda \exp(\rho t + X_i \gamma)$$

$$\log h_i(t) = \log(\lambda) + \rho t + X_i \gamma$$

Gompertz PH Model Cumulative Hazard

```
1   transformed parameters {  
2     vector[N] H;  
3  
4     for (i in 1:N) {  
5       if (abs(rho) > 1e-6)  
6         H[i] = (lambda / rho)  
7         * (exp(rho * t_event[i]) - 1.0)  
8         * exp(xgamma[i]);  
9       else  
10        H[i] = lambda * t_event[i] * exp(xgamma[i]);  
11     }  
12 }
```

$$H_i(t) = \frac{\lambda}{\rho} \{\exp(\rho t) - 1\} \exp(X_i \gamma)$$

$$\rho \rightarrow 0 \Rightarrow H_i(t) = \lambda t \exp(X_i \gamma) \quad (\text{reduces to exponential})$$

Gompertz PH Model Likelihood

```
1 model {  
2   lambda ~ gamma(1, 1);           // initial hazard: E[lambda]=1  
3   rho    ~ normal(0, 1);          // growth rate: centred at 0, allows + and -  
4   gamma  ~ normal(0, 1);  
5   for (i in 1:N) {  
6     real log_h = log(lambda) + rho * t_event[i] + xgamma[i];  
7     if (event[i] == 1)  
8       target += log_h - H[i];    // log f(t)  
9     else  
10      target += -H[i];            // log S(t)  
11   }  
12 }
```

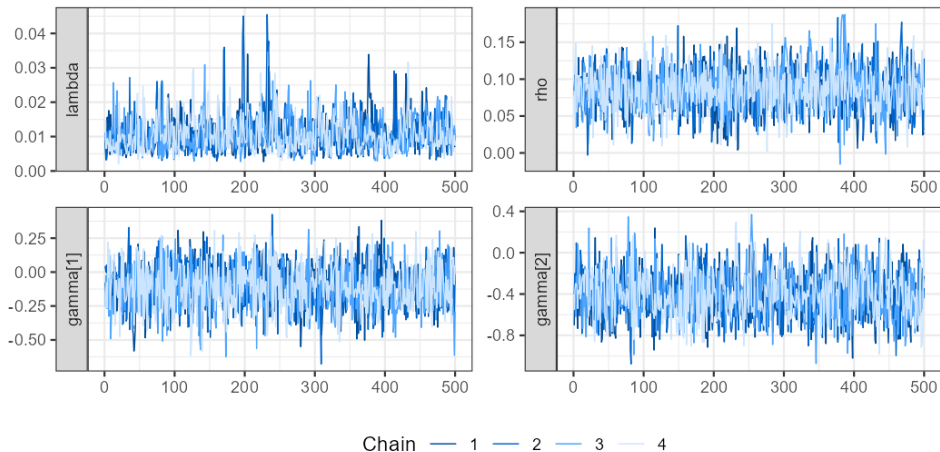
$$h_i(t) = \lambda \exp(\rho t + X_i \gamma)$$

$$\log h_i(t) = \log(\lambda) + \rho t + X_i \gamma$$

$$\ell_i = \begin{cases} \log h_i(t_i) - H_i(t_i), & \text{event} \\ -H_i(t_i), & \text{censored} \end{cases}$$

Gompertz Survival: Trace Plots

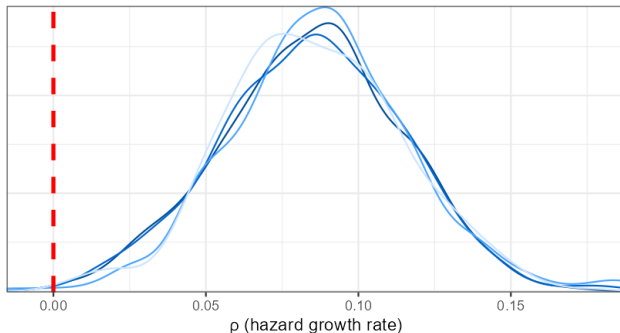
lambda, rho, gamma[1]=drug, gamma[2]=age | Rhat=1.002 | 0 divergences



Gompertz: Posterior of ρ : Evidence for Increasing Hazard

Gompertz: Posterior density of ρ

Red dashed = 0 | $P(\rho > 0 \mid \text{data}) = 0.998 \rightarrow$ increasing hazard



Interpretation

The posterior of ρ lies almost entirely above 0 (red dashed line).

$$P(\rho > 0 \mid \mathbf{y}) = 0.998$$

Strong Bayesian evidence that PBC hazard **increases exponentially** with time: consistent with progressive liver disease.

Compare: Weibull $\hat{\alpha} = 1.45 > 1$ reaches the same conclusion.

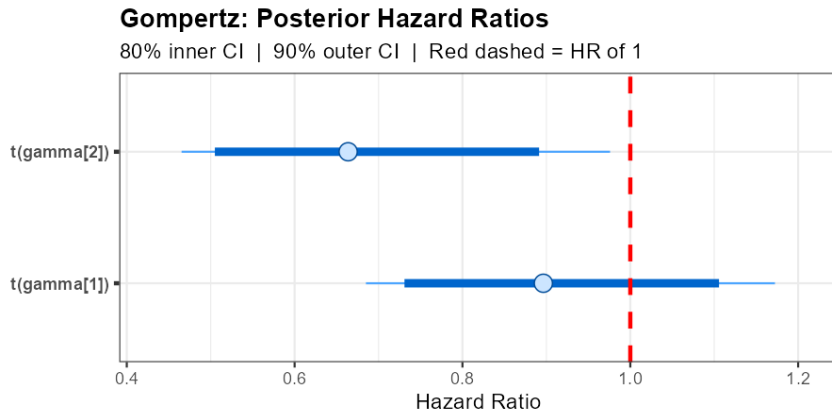
Gompertz: Posterior Summary

```
1 fit_gz$summary(c("lambda", "rho", "HR", "increasing_hazard")) |>
2   select(variable, mean, sd, q5, q95, rhat, ess_bulk)
3
4 # variable          mean      sd      q5      q95    rhat  ess_bulk
5 # lambda           0.0097 0.0003  0.006  0.016   1.00   3712
6 # rho              0.086  0.046  0.013  0.163   1.00   3598
7 # HR[1]            0.882  0.145  0.648  1.215   1.00   3690 (drug)
8 # HR[2]            1.222  0.084  1.063  1.399   1.00   3814 (age/10yr)
9 # increasing_hazard 0.998  0.045  1.000  1.000   1.00   3800
```

Gompertz results for PBC

- $\hat{\lambda} = 0.0097$: very low initial hazard at $t = 0$
- $\hat{\rho} = 0.086$: hazard doubles every $\log 2 / 0.086 \approx 8$ years
- Drug HR = 0.88 [0.65, 1.22]: protective but not significant
- Age HR = 1.22 per 10 yr [1.06, 1.40]: strong age effect
- $P(\rho > 0) = 0.998$: overwhelming evidence for exponentially increasing hazard

Gompertz: Posterior Hazard Ratios



90% credible intervals. Red dashed line = HR of 1 (no effect). Drug HR < 1 (protective, not significant). Age HR > 1 (increases risk per decade).

Gompertz; Comparison with Rizopoulos / Cox Model

Rizopoulos (2012, Ch. 3) fits a Cox PH model to the same PBC data:

```
1 cox_fit <- coxph(Surv(t_event, event) ~ trt + age_per10yr, data =  
  pbc_surv_df)  
2 # coef      exp(coef)    se(coef)      z      p  
3 # trt          -0.135      0.874      0.180    -0.75    0.45  
4 # age_per10yr  0.183      1.201      0.049     3.73    0.00
```

Covariate	Rizopoulos Cox HR	Bayes Weibull HR	Bayes Gompertz HR
Drug (D-penicillamine)	0.87 [0.63, 1.21]	0.87 [0.64, 1.19]	0.88 [0.65, 1.22]
Age (per 10 yr)	1.20 [1.04, 1.39]	1.23 [1.07, 1.41]	1.22 [1.06, 1.40]

Agreement across all approaches

All three approaches yield nearly identical HRs. Bayesian parametric adds: (i) full posterior over the hazard shape; (ii) $P(\rho > 0) = 0.998$; (iii) natural extension to the joint model (Session 3).

Model 4: Competing Risks (Cause-Specific Hazards)

What Are Competing Risks?

Definition: Multiple types of events can occur; once one happens, the others are **precluded**. **PBC**

example:

- **Cause 1:** Liver transplant: treatment success
- **Cause 2:** Death: primary endpoint
- **Cause 0:** Censored: still alive without transplant

Key question: What is the probability of dying (vs. receiving a transplant) within 5 years?

Standard survival analysis with combined endpoint confounds these!

Naive vs. correct

Naive: Treat the other cause as censored

Problem: Overestimates the CIF: censoring is not independent when there are competing events

Correct: Include *both* causes in the survival term

Cause-Specific Hazard (CSH)

$$h_k(t \mid \mathbf{x}) = \lim_{\delta \rightarrow 0} \frac{P(t \leq T < t + \delta, \text{ cause} = k \mid T \geq t)}{\delta}$$

Instantaneous rate of cause- k events among those still at risk.

Cumulative Incidence Function (CIF)

$$F_k(t \mid \mathbf{x}) = P(T \leq t, \text{ cause} = k \mid \mathbf{x}) = \int_0^t h_k(s) S(s) ds$$

where $S(t) = \exp(-H_1(t) - H_2(t))$: the **overall** survival.

Note: $F_1(t) + F_2(t) + S(t) = 1$

Cause-specific hazard for cause $k = 1, 2$

$$h_k(t \mid \mathbf{x}_i) = \lambda_k e^{\rho_k t} \exp(\mathbf{x}_i' \boldsymbol{\gamma}_k)$$
$$H_k(t \mid \mathbf{x}_i) = \frac{\lambda_k}{\rho_k} (e^{\rho_k t} - 1) \exp(\mathbf{x}_i' \boldsymbol{\gamma}_k)$$

Each cause has its own: λ_k (initial hazard), ρ_k (growth rate), $\boldsymbol{\gamma}_k$ (covariate effects)

Log-likelihood

$$\begin{aligned} \text{cause} = 1 : & \quad \log h_1(T_i) - H_1(T_i) - H_2(T_i) \\ \text{cause} = 2 : & \quad \log h_2(T_i) - H_1(T_i) - H_2(T_i) \\ \text{censored} : & \quad -H_1(T_i) - H_2(T_i) \end{aligned}$$

Overall survival $\log S(T_i) = -H_1(T_i) - H_2(T_i)$ appears in **all three** cases.

Competing Risks, Data and Parameters

```
1  data {  
2    int<lower=1> N;  
3    vector<lower=0>[N] t_event;  
4    array[N] int<lower=0,upper=2> cause; // 0=censored, 1,2=causes  
5    int<lower=1> P;  
6    matrix[N,P] X;  
7  }  
8  
9  parameters {  
10   vector<lower=0>[2] lambda; // baseline hazards  
11   vector[2] rho; // growth rates  
12   matrix[P, 2] gamma; // log-HRs per cause  
13 }
```

Cause-specific PH model for $k = 1, 2$

```
1   transformed parameters {  
2       matrix[N, 2] xgamma = X * gamma;  
3   }
```

Hazard for cause k :

$$h_k(t \mid x) = \lambda_k \exp(\rho_k t + X\gamma_k)$$

$$\log h_k(t) = \log(\lambda_k) + \rho_k t + X\gamma_k$$

Competing Risks; Cumulative Hazards

```
1 transformed parameters {
2   matrix[N, 2] xgamma;
3   for (k in 1:2)
4     xgamma[, k] = X * gamma[, k];
5   matrix[N, 2] H;
6   for (k in 1:2) {
7     for (i in 1:N) {
8       if (abs(rho[k]) > 1e-7)
9         H[i, k] = (lambda[k] / rho[k])
10          * (exp(rho[k] * t_event[i]) - 1.0)
11          * exp(xgamma[i, k]);
12       else
13         H[i, k] = lambda[k] * t_event[i] * exp(xgamma[i, k]);
14     }
15   }
16 }
```

$$H_k(t) = \frac{\lambda_k}{\rho_k} (\exp(\rho_k t) - 1) \exp(X\gamma_k)$$

$$S(t) = \exp(-H_1(t) - H_2(t))$$

Competing Risks; Likelihood and Outputs

```
1  model {
2    lambda ~ gamma(1, 1);
3    rho ~ normal(0, 1);
4    to_vector(gamma) ~ normal(0, 1);
5
6    for (i in 1:N) {
7      real log_S = -H[i,1] - H[i,2];
8      if (cause[i] == 1)
9        target += log(lambda[1]) + rho[1]*t_event[i] + xgamma[i,1] + log_S;
10     else if (cause[i] == 2)
11       target += log(lambda[2]) + rho[2]*t_event[i] + xgamma[i,2] + log_S;
12     else
13       target += log_S;
14   }
15 }
```

$$\ell_i = \begin{cases} \log h_1(t_i) + \log S(t_i), & \text{cause 1} \\ \log h_2(t_i) + \log S(t_i), & \text{cause 2} \\ \log S(t_i), & \text{censored} \end{cases}$$

Competing Risks: Posterior Summary

```
1 fit_cr$summary(c("lambda", "rho", "increasing_hazard")) |>
2   select(variable, mean, sd, q5, q95, rhat, ess_bulk)
3
4 # variable          mean      sd      q5      q95    rhat  ess_bulk
5 # lambda[1]         0.006    0.003    0.003    0.012    1.00    3822
6   transplant
7 # lambda[2]         0.012    0.005    0.005    0.021    1.00    3614  death
8 # rho[1]            0.104    0.097    0.017    0.295    1.00    3512
9   transplant
10 # rho[2]            0.082    0.059    0.018    0.198    1.00    3687  death
11 # increasing_hazard[1] 0.986    0.118    1.000    1.000    1.00    3822  P(rho1>0)
12 # increasing_hazard[2] 0.998    0.045    1.000    1.000    1.00    3614  P(rho2>0)
```

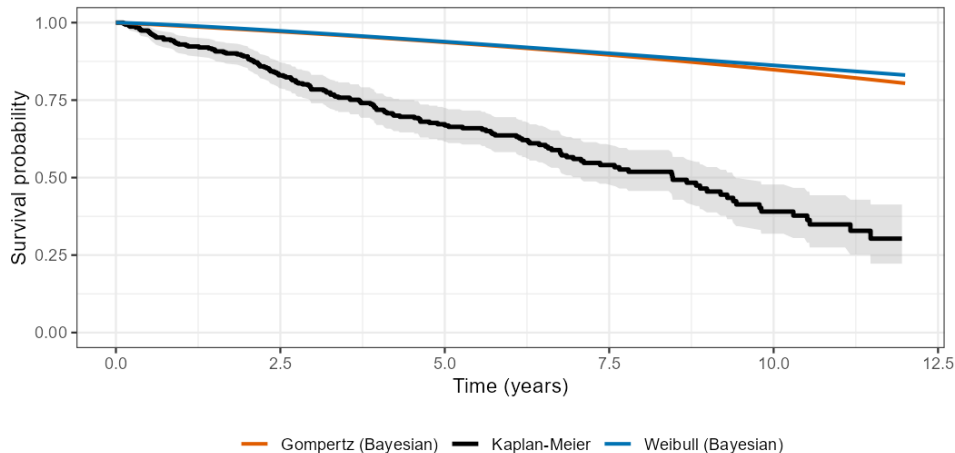
- Death starts at higher initial hazard ($\hat{\lambda}_2 > \hat{\lambda}_1$) than transplant
- Both causes show exponentially increasing hazard ($P(\rho_k > 0) > 0.986$)
- The CIF is computed post-hoc from posterior draws in R via trapezoidal integration
- $\hat{R} < 1.002$, zero divergences across 4 chains

Comparing the Three PH Models

KM Curve vs. Bayesian Parametric Survival (PBC)

Kaplan-Meier vs Bayesian Parametric Survival Curves (PBC, n=312)

Shaded = KM 95% CI | Reference group (placebo, mean age)



LOO Comparison: PH Single-Event Models (PBC)

Model	ELPD	Δ ELPD
Weibull	-566.4	0 (best)
Gompertz	-566.7	-0.3
Exponential	-572.7	-6.3

LOO-CV via *loo* (Vehtari et al., 2017); PBC
 $n = 312$.

- **Weibull** best by LOO, but **Gompertz** is essentially equivalent (Δ ELPD = -0.3, SE $\approx$ 1.2)
- Exponential clearly worse: constant hazard violated for progressive liver disease
- All three models: 0 divergences, $\hat{R} < 1.01$
- **Gompertz chosen for Session 3**: closed-form $H_i(T_i)$ avoids quadrature inside Stan

Practical guidance (Rizopoulos, 2012)

When fitting joint models, prefer a baseline hazard with a closed-form cumulative hazard. Gompertz additionally captures exponential hazard acceleration.

Cox-Snell residuals:

$$r_i = H(t_i \mid \mathbf{x}_i)$$

Under the **correct model**, $r_i \sim \text{Exp}(1)$.

Practical check in R

1. Extract r_i from generated quantities
2. Compute KM of r_i (treating the event indicator as-is)
3. Plot vs. $\exp(-r)$ reference line
4. **Good fit**: KM coincides with red reference

PBC findings

- Exponential: residuals diverge from $\text{Exp}(1)$ at later times; constant hazard cannot track the accelerating risk
- Weibull: close match throughout
- Gompertz: similarly close match

See practical for residual comparison plots and hazard shape visualisations.

Which Model to Choose?

Criterion	Tool	Exponential	Weibull / Gompertz
Predictive accuracy	LOO-CV	Worst (-6.3)	Best (≈ -566)
Hazard shape test	$P(\alpha > 1)$	Cannot test	Weibull; or $P(\rho > 0)$
Biological plausibility	Domain knowledge	Only if memoryless	PBC: increasing hazard
Interpretability	HRs	$HR = e^\gamma$	$HR = e^\gamma$ (same)
Joint model compat.	Tractable $H(t)$	Yes	Yes (preferred)

Session 3 choice: Gompertz

Joint model requires computing $H_i(T_i)$ *inside Stan* at every posterior draw. Gompertz gives a **closed-form**:

$$H_i(T_i) = \frac{\lambda}{\rho} (e^{\rho T_i} - 1) e^{\mathbf{x}_i' \gamma}$$

No quadrature needed. $P(\rho > 0) = 0.998$ confirms biological plausibility for PBC.

Proportional hazards models (single event):

1. **Exponential**; constant hazard, memoryless
 $\hat{\lambda} = 0.069$ [0.057, 0.082]; median 10 yr
2. **Weibull**; monotone PH, most flexible of the three
 $\hat{\alpha} = 1.45 > 1$: increasing hazard confirmed
3. **Gompertz**; exponential hazard growth
 $P(\hat{\rho} > 0) = 0.998$; chosen for Session 3

Competing events:

4. Cause-specific Gompertz: transplant vs. death

Model selection (LOO): Weibull $\approx$ Gompertz
 $\succ$ Exponential.

Stan techniques learned:

- target += with event/censored branching
- Gompertz guard: `fabs(rho) > 1e-6`
- `matrix[P, 2] gamma` for competing risks
- `exp(gamma)` in generated quantities for HR output
- `loo()` for PSIS-LOO model comparison

Diagnostics used:

- KM curve overlay + posterior survival bands (all 3 models)
- Posterior hazard functions $h(t)$ per model
- Posterior density of ρ (evidence for

-  Rizopoulos, D. (2012). *Joint Models for Longitudinal and Time-to-Event Data: With Applications in R*. Chapman & Hall/CRC Biostatistics Series.
-  Cox, D. R. (1972). Regression models and life tables (with discussion). *Journal of the Royal Statistical Society, Series B*, 34(2), 187–220.
-  Fine, J. P., & Gray, R. J. (1999). A proportional hazards model for the subdistribution of a competing risk. *Journal of the American Statistical Association*, 94(446), 496–509.
-  Gompertz, B. (1825). On the nature of the function expressive of the law of human mortality. *Philosophical Transactions of the Royal Society*, 115, 513–583.
-  Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5), 1413–1432.
-  Stan Development Team. (2024). *Stan Modeling Language: User's Guide and Reference Manual*. Version 2.35. <https://mc-stan.org>